

CLINICAL INVESTIGATION

Single-Masked Randomized Phase 2 Study Assessing 2 Forms of Hypofractionated Proton Therapy in Patients With Large Choroidal Melanomas



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Purpose: Patients with large uveal melanomas are at major risk of liver metastases. Some patients are reluctant to undergo the standard treatment (ie, immediate enucleation). Proton therapy yields 5-year local control rates and eyeball retention of >85% and ≈20% in large uveal melanomas. Patients with T3/T4 uveal melanomas refusing enucleation were randomized between standard 4 to 13 Gy-fraction or moderately hypofractionated 8 to 6.5 Gy-fraction proton therapy. The main endpoint was the 2-year local recurrence-free survival without enucleation.

Methods and Materials: A single-masked 1:2 randomized phase 2 trial was conducted between 2015 and 2017 with planned endoresection and distance to the posterior pole as strata. Local events were defined as local relapse, or enucleation due to complications or relapse.

Results: The 32 patients, with a mean age of 64 years, had T3/4 (N = 17/15), M1 (N = 2) uveal melanomas, of mean tumor diameter and thickness of 16.5 mm and 9.1 mm, and of posterior location in 56.5%. Median follow-up was 56.7 months. The 2-year local recurrence-free survival rate without enucleation was 79% (95% confidence interval, 65%-96%), similar in both arms. There were 9 enucleations, 3 at relapse and 6 for toxicities. Twelve patients had distant metastases. The 2-year-overall survival was 72% (95% confidence interval, 58%-89%). At baseline, visual acuity by average logarithm value of the minimum angle of resolution was 0.68 and 0.70 in the standard and experimental arms, and at last follow-up 2 and 1.7, with mean differences of 1.44 and 1.01, respectively ($P = .39$).

Conclusion: An 8-times 6.5 Gy scheme is feasible without deteriorating local control and with similar toxicity rates in patients with large uveal melanomas. Larger studies incorporating adjuvant treatments are warranted. © 2023 Elsevier Inc. All rights reserved.

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